Affordability of Houses in the Netherlands

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Programming for Economists

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1 1 Social Problem

Housing affordability has become an increasingly important social issue in the Netherlands over the past decade. House prices have risen substantially in many parts of the country, while income growth have not kept pace. As a result, many households, particularly first-time buyers and younger generations, face greater difficulties entering the housing market, according to DNB (DNB, 2026). In addition, CBS states that housing costs in the Netherlands are higher than most EU countries (CBS, 2024).

Understanding whether house prices have increased faster than incomes is therefore important for evaluating changes in housing affordability. If housing prices grow more rapidly than household incomes, purchasing a home becomes increasingly difficult for a large share of the population. This may have significant social and economic consequences, including increased inequality between homeowners and non-homeowners.

The main research question of this study is:

Have housing prices in the Netherlands increased faster than incomes over the past ten years?

To answer this question, the following sub-questions are examined:

1. How have house prices and average incomes developed between 2014 and 2024?
2. How has housing affordability changed during this period?
3. Are there significant differences in housing affordability between Dutch provinces?
4. Is there a visible change in the affordability trend after the increase in mortgage interest rates around 2022?

2 Part 2 - Data Sourcing

2.1 2.1 Data Loading

The data were collected from Statistics Netherlands (CBS) using the `cbsodataR` package in R. The main dataset used is CBS table 70072NED, which contains regional socio-economic and housing indicators. After downloading the dataset, label columns were added and a local copy was saved to make the analysis reproducible. The spatial data for the province map were loaded using CBS spatial boundary data.

2.2 2.2 Dataset Summary

The primary data for this study is sourced from Statistics Netherlands (CBS). The original dataset contains 24,335 observations and 326 variables, including socio-economic indicators, average employee income, WOZ property values, regional classifications, and demographic variables.

For the analysis, two main subsets are created from the CBS dataset. The first subset contains provincial income data, while the second contains national WOZ values over time. These subsets are cleaned separately and then merged by year and region. This allows the analysis to focus on the relationship between house values and income while retaining the regional structure needed to compare affordability across Dutch provinces and sub-populations.

Year	RegioS	RegioS_label	GemiddeldeWOZWaardeVanWoningen_98
2023	PV20	Groningen (PV)	268
2023	PV21	Fryslân (PV)	277
2023	PV22	Drenthe (PV)	294
2023	PV23	Overijssel (PV)	325
2023	PV24	Flevoland (PV)	347
2023	PV25	Gelderland (PV)	364
2023	PV26	Utrecht (PV)	447
2023	PV27	Noord-Holland (PV)	460

Year	RegioS	RegioS_label	GemiddeldeWOZWaardeVanWoningen_98
2023	PV28	Zuid-Holland (PV)	359
2023	PV29	Zeeland (PV)	281
2023	PV30	Noord-Brabant (PV)	370
2023	PV31	Limburg (PV)	288

Year	RegioS	RegioS_label	BronInkomenAlsWerknemer_141
2023	PV20	Groningen (PV)	37.5
2023	PV21	Fryslân (PV)	37.9
2023	PV22	Drenthe (PV)	39.7
2023	PV23	Overijssel (PV)	39.5
2023	PV24	Flevoland (PV)	40.4
2023	PV25	Gelderland (PV)	40.6
2023	PV26	Utrecht (PV)	44.4
2023	PV27	Noord-Holland (PV)	44.7
2023	PV28	Zuid-Holland (PV)	42.4
2023	PV29	Zeeland (PV)	40.3
2023	PV30	Noord-Brabant (PV)	41.8
2023	PV31	Limburg (PV)	39.9

Year	RegioS	RegioS_label	GemiddeldeWOZ	InkomenWerknemer	statcode
2023	PV20	Groningen (PV)	268	37.5	PV20
2023	PV21	Fryslân (PV)	277	37.9	PV21
2023	PV22	Drenthe (PV)	294	39.7	PV22
2023	PV23	Overijssel (PV)	325	39.5	PV23
2023	PV24	Flevoland (PV)	347	40.4	PV24
2023	PV25	Gelderland (PV)	364	40.6	PV25
2023	PV26	Utrecht (PV)	447	44.4	PV26
2023	PV27	Noord-Holland (PV)	460	44.7	PV27
2023	PV28	Zuid-Holland (PV)	359	42.4	PV28
2023	PV29	Zeeland (PV)	281	40.3	PV29
2023	PV30	Noord-Brabant (PV)	370	41.8	PV30
2023	PV31	Limburg (PV)	288	39.9	PV31

Table 4: Descriptive statistics for home value in provinces in the Netherlands, 2023

Year	RegioS_label	InkomenWerknemer	GemiddeldeWOZ
2015	Nederland	30.3	206
2016	Nederland	31.4	208
2017	Nederland	32.0	217
2018	Nederland	32.3	230
2019	Nederland	33.7	250
2020	Nederland	35.3	271
2021	Nederland	37.1	290
2022	Nederland	39.0	317
2023	Nederland	41.8	368
2024	Nederland	44.7	378

Year	RegioS_label	InkomenWerknemer	GemiddeldeWOZ

Variable	Mean	Min	Max	SD
Average WOZ Value	340.00	268.0	460.0	64.25
Employee Income	40.76	37.5	44.7	2.24

2.3 2.3 Variables

The variables included in this study predominantly capture Socio-Economic Status (SES) and housing market characteristics. The key indicators can be broken down into three main categories: 1. SES metrics: Disposable income per employee serves as our primary continuous of household financial capacity. 2. Market metrics: The WOZ property value represents the official appraised market valuation of resedential real estate over time. 3. Demographic metrics: The use of regions and urbanization levels allow for granular sub-group analysis.

Rather than being collected through self-supported individual interviews or surveys, this dataset consist entirely of administrative data collected by Statistics Netherlands (CBS). The income and demographic data originate directly from national municipal registries (BRP) and tax authority records, while the housing data is compiled from the national valuation registry (Landelijke Voorziening WOZ).Because this data relies on offical administrative registration rather than sample surveys, it minimizes self-reporting bias and provides comprehensive coverage of the target populations.

3 Part 3 - Quantifying

3.1 3.1 Data cleaning

To prepare our data for quantitative analysis, we executed several structured cleaning steps. First, provincial location data was originally recorded in numerical codes. We mapped and translated these administrative codes into their respective official Dutch province names. Second, we systematically handled missing values (NAs). To safeguard the statistical integrity of our analysis, we selected a time period that is relevant and without any missing values (NAs) at other variables.

3.2 3.2 Generate necessary variables

The affordability ratio divides the average WOZ value of homes by average employee income. Because both variables are measured in thousands of euros, the ratio can be interpreted as how many times the income of a worker corresponds to their house value, on average. A higher ratio indicates lower affordability.

Table 6: Housing Affordability by Province, 2023

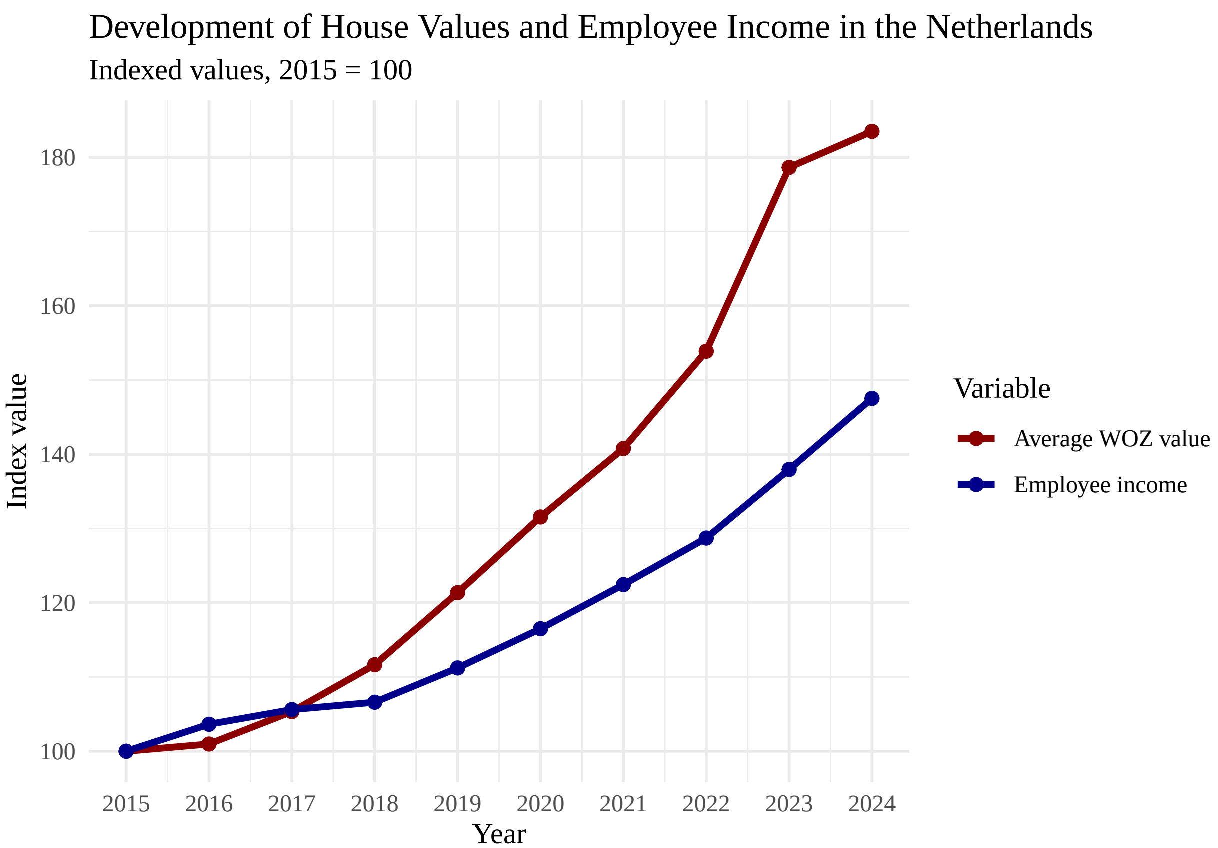
Province	WOZ	Income	Affordability
Groningen (PV)	268	37.5	7.15
Fryslân (PV)	277	37.9	7.31
Drenthe (PV)	294	39.7	7.41
Overijssel (PV)	325	39.5	8.23
Flevoland (PV)	347	40.4	8.59
Gelderland (PV)	364	40.6	8.97
Utrecht (PV)	447	44.4	10.07
Noord-Holland (PV)	460	44.7	10.29
Zuid-Holland (PV)	359	42.4	8.47
Zeeland (PV)	281	40.3	6.97
Noord-Brabant (PV)	370	41.8	8.85

Province	WOZ	Income	Affordability
Limburg (PV)	288	39.9	7.22

Table 7: National Affordability, 2015-2024

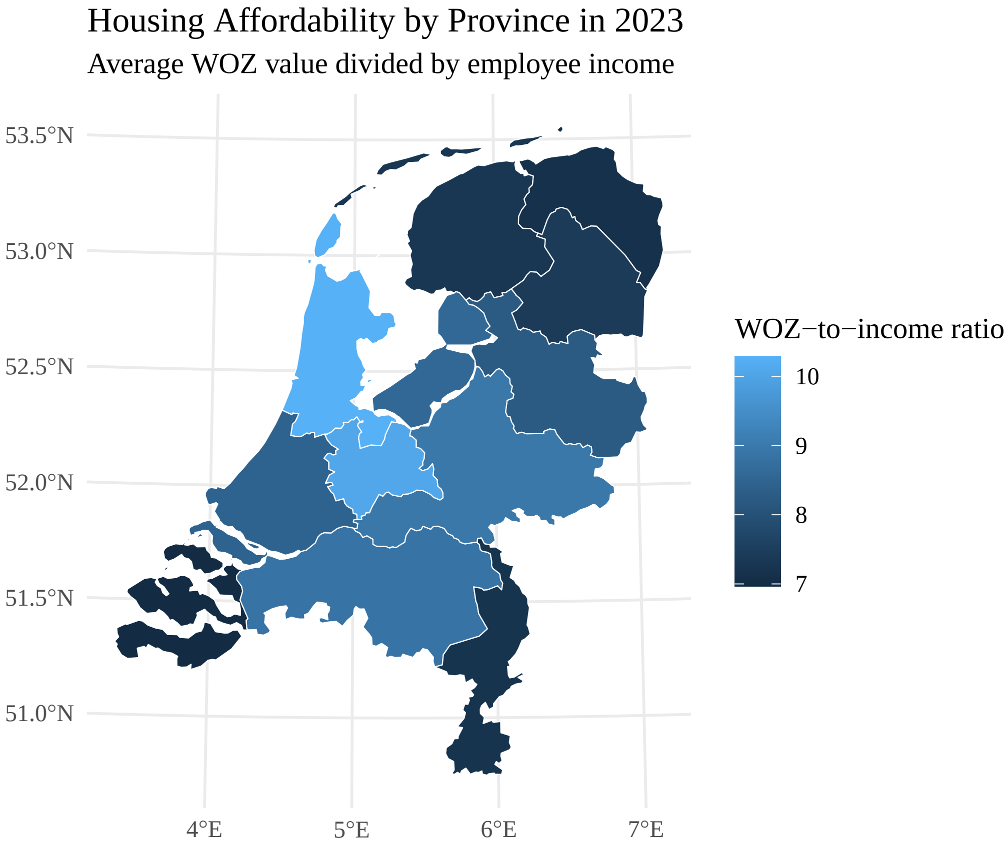
Year	Income	WOZ	Affordability	WOZ_growth	Income_growth	Growth_gap
2015	30.3	206	6.798680	NA	NA	NA
2016	31.4	208	6.624204	0.9708738	3.630363	-2.659489
2017	32.0	217	6.781250	4.3269231	1.910828	2.416095
2018	32.3	230	7.120743	5.9907834	0.937500	5.053283
2019	33.7	250	7.418398	8.6956522	4.334365	4.361287
2020	35.3	271	7.677054	8.4000000	4.747775	3.652226
2021	37.1	290	7.816712	7.0110701	5.099150	1.911920
2022	39.0	317	8.128205	9.3103448	5.121294	4.189051
2023	41.8	368	8.803828	16.0883281	7.179487	8.908841
2024	44.7	378	8.456376	2.7173913	6.937799	-4.220408

3.3 3.3 Temporal Variation



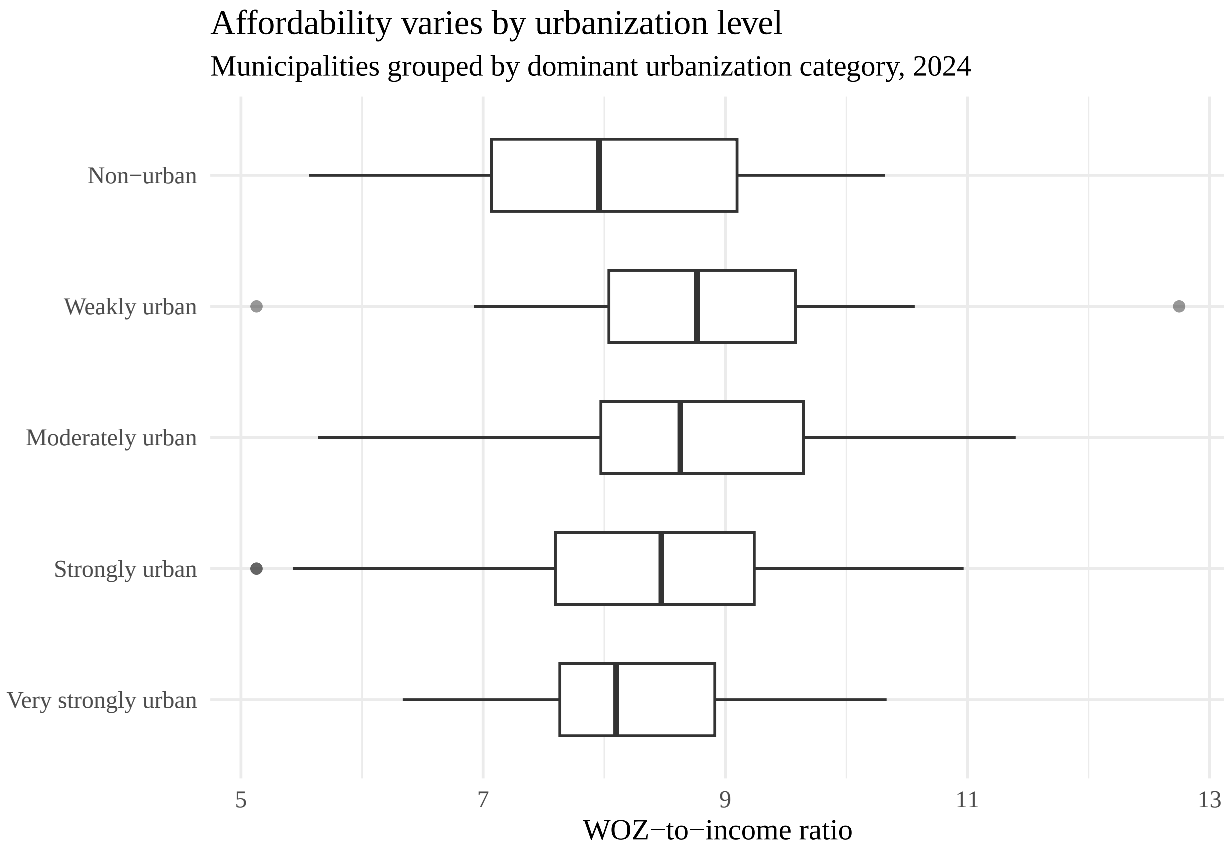
3.4 3.4 Spatial Variation

The map visualizes how housing affordability differs across Dutch provinces in 2023. The WOZ-to-income ratio is highest in Noord-Holland and Utrecht, meaning that average house values are high relative to employee income in these provinces. Provinces such as Zeeland, Groningen and Limburg show lower ratios, suggesting relatively better affordability. This spatial pattern is relevant because the housing affordability problem is not evenly distributed across the Netherlands. However, the province-level this map does not show the differences inside provinces which could vary a lot. For example, inside cities the affordability is likely different than outside cities.



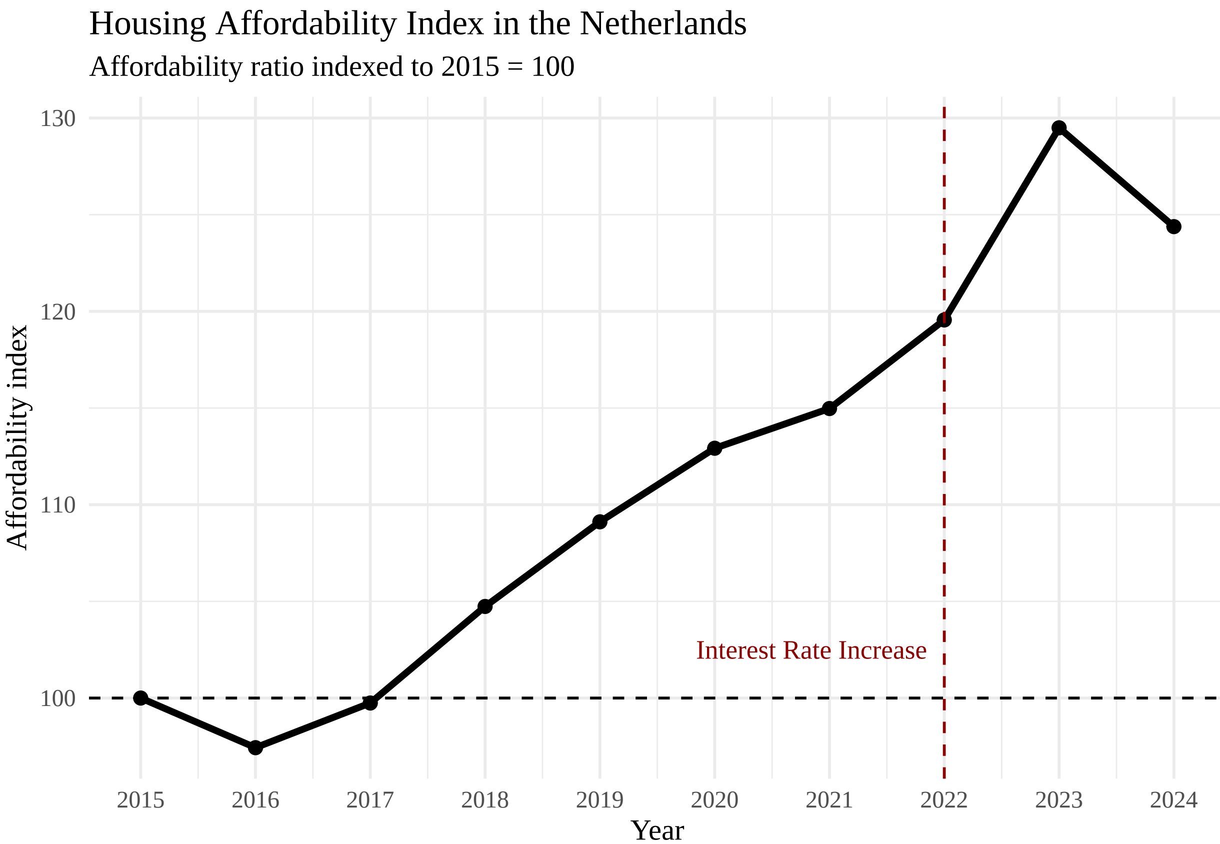
3.5 3.5 Sub-Population Variation

The boxplot compares housing affordability across municipalities grouped by the level of urbanization. This is relevant because housing pressure is often expected to be stronger in more urbanized areas, where demand for housing is higher and land is more scarce. The plot shows that affordability ratios differ across urbanization groups, but also that there is substantial variation within each group. This suggests that urbanization matters, but it does not fully explain differences in affordability.



3.6 3.6 Event Analysis

Interest rates are included as an external event because mortgage financing costs affect housing demand and house prices. Harris (1989) argues that housing prices are influenced by the real cost of mortgage financing, where nominal mortgage rates interact with expectations of future house-price appreciation. This supports using the 2022 increase in mortgage interest rates as a relevant event for analyzing changes in the WOZ-to-income affordability index. In this assignment, 2022 is treated as the turning point because mortgage interest rates started increasing after a period of decreasing rates (De Nederlandsche Bank, 2026). If higher interest rates reduce borrowing capacity and cool housing demand, we would expect WOZ growth to slow relative to income growth after this point. This pattern is visible in the affordability index: affordability worsened until 2023, but improved slightly in 2024 as income growth exceeded WOZ growth.



The effect of higher interest rates is not expected to appear immediately in the affordability index. Housing markets adjust with a delay because buyers and sellers need time to change their behavior. Therefore, the affordability index can still increase after the interest-rate turning point in 2022. This helps explain why affordability continues to deteriorate until 2023, before improving slightly in 2024. However, this event analysis remains descriptive. Since there is no control group and only a short post-event period, the graph cannot prove that higher interest rates caused the change in affordability. It only shows that the affordability trend changed after the interest-rate environment shifted.

4 Part 4 - Discussion

4.1 4.1 Findings

4.1.1 Affordability ratio

The affordability ratio shows that housing affordability worsened between 2015 and 2024. In 2015, the national affordability ratio was approximately 6.80, meaning that the average WOZ value was about 6.8 times average employee income. By 2023, this ratio had increased to approximately 8.80. Although the ratio declined slightly in 2024, it remained clearly above the 2015 level. This suggests that housing values increased faster than employee income over most of the period.

4.1.2 Growth gap

The year-on-year growth gap shows whether WOZ values grew faster than employee income in a given year. Positive values indicate years in which housing became less affordable because house values increased faster than income. The largest positive growth gap occurred in 2023, when WOZ growth exceeded income growth by a large amount. In 2024, the growth gap became negative, meaning that income growth exceeded WOZ growth.

4.1.3 Temporal variation

The temporal analysis shows that WOZ values and employee income initially moved relatively closely together. After 2017, however, the indexed WOZ value increased much faster than indexed employee income. This divergence reached its highest point in 2023. Although in 2024 the gap seems to decrease, this was not enough to counteract the gaps growth of previous years.

4.1.4 Spatial variation

The spatial analysis shows that housing affordability differs strongly across Dutch provinces. Noord-Holland and Utrecht have the highest WOZ-to-income ratios, which means that average house values are high relative to employee income in these provinces. Zeeland, Groningen and Limburg show lower ratios, suggesting relatively better affordability. However, province-level averages should be interpreted carefully because they hide variation within provinces, especially between large cities and smaller municipalities.

4.1.5 Sub-population variation

The urbanization-level analysis shows that affordability differs between types of municipalities. More urbanized areas generally show higher affordability pressure, but there is also substantial variation within each urbanization category. This means that urbanization is relevant, but it does not fully explain housing affordability differences. Other factors may influence the affordability of houses, like for example commuting possibilities that are available in a certain place.

4.1.6 Event

The event analysis suggests that the affordability index continued to worsen after interest rates started increasing in 2022, but improved slightly in 2024. This delayed pattern is expected because housing markets do not adjust immediately to interest-rate changes. Buyers and sellers need time to adjust, and WOZ values may change with a lag. The analysis however, should not be seen as causal proof. The report does not include a control group, and other developments may also have influenced the trend.

4.2 Conclusion

Overall, the analysis suggests that Dutch housing affordability became worse between 2015 and 2024. Average WOZ values increased substantially faster than average employee income. The problem is not evenly distributed across the Netherlands: affordability pressure appears strongest in provinces such as Noord-Holland and Utrecht and differs across urbanization levels. The slight improvement in 2024 is important, but it does not undo the broader deterioration observed over the full period.

5 Part 5 - Reproducibility

5.1 5.1 Github repository link

https://github.com/JulesBierings/PFE-group6-7-paper-2026

6 References

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